Project Details

Project Name:
AI-Based Emergency Room Patient Flow Prediction

Meeting Date:
16-08-2026

Meeting Time:
11:45 – 12:20

Mentor:
Mr. Parag Tamgadge

Meeting Number:
3

1. Members Attended

Name	Role	Present (Y/N)
Mahika Mangaonkar	Team Leader / Chatbot Development	Y
Smurti Rani Khadanga	Data Preprocessing & Data Analysis	Y
Aparupa Samal	ML Model Development	Y
Anu Kumari	Dataset Generation & ML Model Development	Y

2. Discussion Points

A. Problem Statement

- The team discussed the problem of predicting Emergency Room patient flow, patient arrivals, and related operational patterns using historical and generated healthcare data.
- The objective is to develop machine learning models that can help identify patient flow patterns and support better emergency department planning and decision-making.
- The current focus is to move from the data preparation stage towards model development and training.
- A key requirement discussed was to begin training the respective models and make measurable progress by the end of the current week.

B. Proposed Solution

- The team discussed developing an AI-based Emergency Room Patient Flow Prediction system using machine learning models trained on relevant patient-flow data.
- The solution will involve multiple models/components addressing the different prediction requirements of the project.
- A synthetic dataset has been created to supplement the available data and provide sufficient data for initial model development and experimentation.
- Anu's XGBoost model will also be used as an input/supporting component for the chatbot so that the chatbot can provide responses based on the project's prediction outputs.
- The agreed approach is to proceed with training and evaluating the respective models before moving further with complete system integration.

Alternatives Considered:

- Different ML approaches and models can be evaluated during the training and experimentation phase based on their performance on the available dataset.

Final Approach Agreed Upon:

- Proceed with model training using the prepared datasets.
- Evaluate the respective models and identify suitable models for further integration.
- Use the output of Anu's XGBoost model as part of the chatbot's functionality.

C. Implementation Progress

Team Member	Contribution	Current Status	Remarks
Mahika Mangaonkar	Chatbot development and project coordination	In Progress	Will integrate the chatbot with the model outputs; specifically, Anu's XGBoost model will be used to support chatbot responses.
Smurti Rani Khadanga	Dataset cleaning and preprocessing	Completed	Cleaned and prepared the available dataset for further model development.
Aparupa Samal	Machine learning model research and development	In Progress	Will proceed with training the assigned model using the prepared dataset.

Team Member	Contribution	Current Status	Remarks
Anu Kumari	Synthetic dataset generation and XGBoost model development	In Progress	Created the synthetic dataset and developed the XGBoost model to be further used in the project.

Overall Progress Summary

Completed:

- Initial project planning and problem understanding.
- Dataset cleaning and preprocessing.
- Synthetic dataset creation by Anu.
- Initial XGBoost model development.

In Progress:

- Training and evaluation of the respective ML models.
- Further development of the XGBoost model.
- Planning chatbot integration with the model outputs.

Pending:

- Completion and evaluation of all respective ML models.
- Integration of trained models with the main application.
- Integration of the chatbot with the trained model outputs.
- End-to-end testing and validation.

Risks/Dependencies:

- Model performance depends on the quality and suitability of the available datasets.
- Chatbot development is dependent on having stable and usable outputs from the trained ML models.
- Further model selection may depend on comparative evaluation results.

3. Suggestions Provided by Mentor

Since the mentor did not provide a specific change in the overall project direction, I would keep this section factual:

- Begin working on the training of the respective ML models and make progress by the end of the week.
- Focus on understanding the implementation and underlying concepts of the models being used.
- The team should be prepared to explain the theoretical concepts related to the models and technologies used in the project during the upcoming mentoring session.
- Technical improvements: Complete model training and evaluate the performance of the respective models.
- Documentation improvements: Maintain proper documentation of model development, training, and evaluation.
- Research/validation required: Compare model performance using appropriate evaluation metrics.
- Other observations: The chatbot should make use of the trained model outputs; Anu's XGBoost model will be used as the initial model for chatbot integration.

4. Actions for the Next Week

Task	Assigned To	Expected Outcome	Target Date
Train the assigned ML models using the prepared datasets	All relevant team members	Initial trained models with performance results	End of this week
Further evaluate the XGBoost model	Anu Kumari	Evaluated XGBoost model and prediction outputs	End of this week
Begin chatbot integration with the XGBoost model	Mahika Mangaonkar	Initial chatbot-model integration	End of this week
Continue model research and evaluation	Aparupa Samal	Comparison of suitable ML approaches/models	End of this week
Validate and prepare datasets required for model training	Smurti Rani Khadanga	Training-ready datasets	End of this week
Study theoretical concepts related to the implemented models	Entire Team	Preparation for theory-based questions in the next mentor session	Before next meeting

Goals for Next Meeting

- Goal 1: Have the respective ML models trained with initial evaluation results.
- Goal 2: Have the chatbot connected to or tested with the outputs of Anu's XGBoost model.
- Goal 3: Be prepared to explain the theoretical concepts, working principles, and implementation choices of the models used in the project.

Prepared By:
Mahika Mangaonkar

Reviewed By (Mentor):

Mr. Parag Tamgadge

Date:
16-08-2026